INCH-POUND
MIL-DTL-24643/21F
1 October 2009
SUPERSEDING
MIL-DTL-24643/21E
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +105 °C, TYPES LSTCJU AND LSTCTU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Watertight with Circuit Integrity

First	-	One iron and one constantan conductor in Type LSTCTU, for Type J thermocouple in accordance with ANSI MC96.1; one copper and one constantan in Type LSTCTU, for Type T thermocouple in accordance with ANSI MC96.1 (see table I for diameter).
Second	-	Thermoset insulation (see table I for diameter).
Third	-	Optional glass braid.
Fourth	-	Covering. Printed identification as follows: Type LSTCJU shall have the legend “8 (GRAY)” printed on the iron conductor and the legend “3 (RED)” printed on the constantan conductor. Type LSTCTU shall have the legend “6 (BLUE)” printed on the iron conductor and the legend “3 (RED)” printed on the constantan conductor (see table I for diameter).
Fifth	-	The two conductors shall be cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm, well-rounded assembly.
Sixth	-	An optional binder.
Seventh	-	Cross-linked polyolefin jacket. Minimum average wall thickness of 0.040 inch.

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TABLE I. Details.

Military part no. M24643/21	Type and size	Numbers of strands and strand diameter (inch)	Diameter over insulation min. avg. (inch)	Diameter over covering max. (inch)	Overall diameter	
					min. (inch)	max. (inch)
-01UN	LSTCJU-4	7/0.025	0.105	0.140	0.400	0.430
-02UN	LSTCTU-4	7/0.025	0.105	0.140	0.400	0.430

REQUIREMENTS:

Qualification required.

INSPECTIONS:

Basic Electricals:

Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	3000
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	500
ANSI MC96.1 calibration limits of error	
Type LSTCJU	±2.2 °C
Type LSTCTU	±1 °C
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure
Abrasion resistance, extruded insulation only (scrapes min.)	250
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Crack resistance, extruded insulation only	No damage

Group B:

Cold bending cable	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Drip (95±1 °C)	Zero
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700

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Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Gas flame (1 hour)	No failure
Group C:	
Physicals (aged) air oven	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Cold working (minus 20±2 °C)	No damage
Gas flame (3 hours)	No failure
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2

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Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSTCJU-4 and LSTCTU-4	2000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-025)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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